INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

QUADRANT, FIRE CONTROL - M18A1

Reactivated after 16 August 2012 and may be used for new and existing designs and acquisitions

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Quadrant, Fire Control: M18A1, which provides a measure of elevation angles for 155mm towed howitzers. It is equipped with numerical counters which indicate elevation settings and corrections.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 and 4 of this specification, whether or not documents are listed in this section.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or emailed to picaardecstdznbranch@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-F-13926 - Fire Control Materiel, Manufacture, and Inspection, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of federal and military specifications, standards, and handbooks are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

11747195	-	Vibration and/or Shock Fixture for Quadrant Fire Control XM17 and/or XM18
11747969	-	Gage, Interchangeability Max
13005102	-	M18A1 Quadrant
13010955	-	Plate, Identification

(Copies of these drawings may be requested online at pica.drawing.request@conus.army.mil or from U.S. Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny Arsenal, NJ 07806-5000)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.3 Quadrant. The quadrant shall conform to drawing 13005102.

3.4 Environmental.

3.4.1 Storage temperature. The quadrant shall be thermally stabilized at temperatures of +160 ± 5°F and -60 ± 5°F for a period of not less than 4 hours at each temperature. While thermally stabilized at these temperatures, the quadrant shall be inspected and shall not exhibit any signs of physical failure or damage. Upon return to and stabilization at standard ambient temperature, +60°F to +90°F, the quadrant shall meet the requirements of 3.5 through 3.14 inclusive.

3.4.2 Operating temperature. The quadrant shall meet the requirements of 3.8 through 3.14 inclusive while exposed to and thermally stabilized at temperatures of +145 ± 5°F and -50 ± 5°F for a period of not less than 4 hours at each temperature. Upon return to and stabilization at standard ambient temperature, +60°F to +90°F, the quadrant shall meet the requirements of 3.5 through 3.14 inclusive.

3.4.3 Shock. The quadrant shall withstand a total of 75 shock impulses along 3 mutually perpendicular axes as defined in Figure 1. Each shock impulse shall be a half sine wave with a time duration of .010 ± .001 seconds and peak amplitude for each shock impulse as shown in Table I. Subsequent to shock, the quadrant shall show no evidence of damage or physical failure and shall meet the requirements of 3.5 through 3.14 inclusive.

TABLE I. Shock orientation and impulse.

Orientation	Shock impulse and direction	Number of impulses
Longitudinal	-90 g's, +90 g's	15 each direction
Vertical	-80 g's, +80 g's	15 each direction
Transverse	-80 g's	15 total

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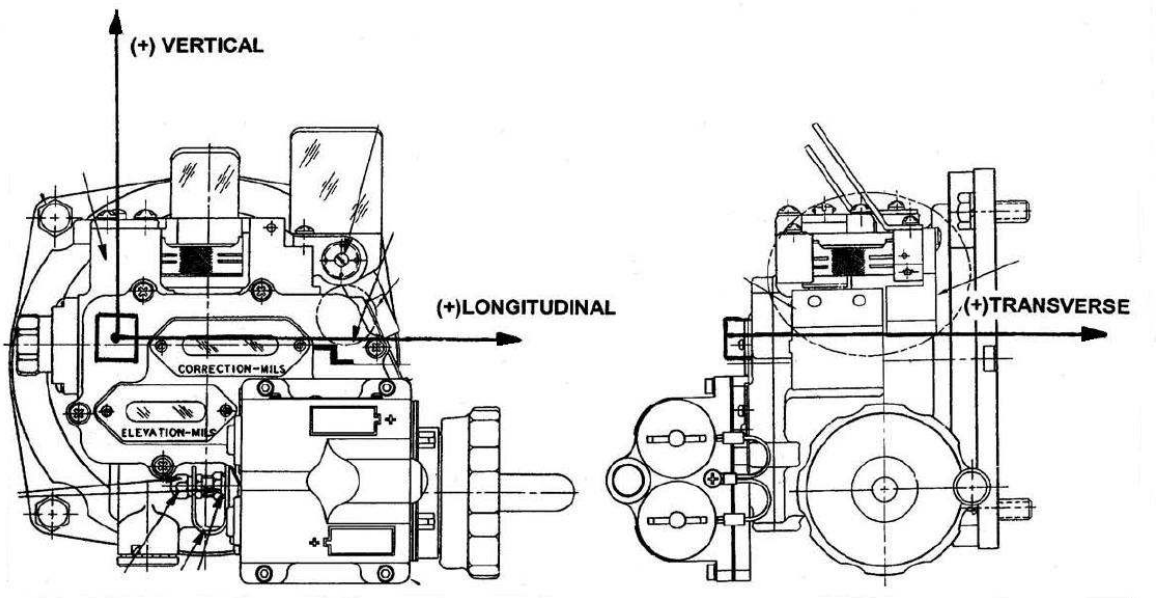


FIGURE 1. Orientation of shock loads.

3.4.4 Vibration "A". The quadrant shall withstand a total of 270 minutes of sweep-cycle vibration. The vibration shall be applied for 90 minutes along each of the three mutually perpendicular major axes shown in Figure 1. A complete sweep-cycle shall consist of vibration from origin (5 Hz at 1 inch double amplitude) to mid-point (5 ± 0.5 g's at 500 Hz) to origin, and shall have a duration of 15 ± 1 minutes. Double amplitude shall be constant at 1 inch between 5 Hz and 10 Hz, and varied with frequency to maintain a constant 5 ± 0.5 g's acceleration between 10 Hz and 500 Hz. Upon completion of vibration, the quadrant shall exhibit no evidence of damage or physical failure and shall meet the requirements of 3.5 through 3.14 inclusive.

3.4.5 Vibration "B". The quadrant shall be vibrated in the vertical plane shown in Figure 1 for a total of 15 ± 1 minutes of sweep cycle vibration. A complete sweep-cycle shall consist of vibration from origin (5 Hz at 1 inch double amplitude) to mid-point (5 ± 0.5 g's at 500 Hz) to origin, and shall have a duration of 15 ± 1 minutes. Double amplitude shall be constant at 1 inch between 5 Hz and 10 Hz and varied with frequency to maintain a constant 5 ± 0.5 g's acceleration between 10 Hz and 500 Hz. Upon completion of vibration, the quadrant shall exhibit no evidence of damage or physical failure and shall meet the requirements of 3.5 through 3.14 inclusive.

3.5 Sealing. The quadrant shall be capable of sustaining an internal pressure of 5.0 ± 0.10 pounds per square inch gage (psig) for a minimum of one hour with no evidence of leakage.

3.6 Purging. The interior of the quadrant shall be purged with dry nitrogen supplied at a pressure of 8.00 ± 0.25 psig until the dew point of the emergent gas is no higher than -25°F. Purging apparatus shall be removed, and subsequent to the seating of the pressure relief valve, the quadrant shall be pressured to 1.0 ± 0.1 psig for all testing and delivery.

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3.7 Cleanliness. There shall be no evidence of moisture, grease, fingerprints, condensates, dirt, debris, scratches, fractures or adhesive separations on glass windows or counters.

3.8 Performance.

3.8.1 Elevation counter excursion range. The elevation counter shall cover the excursion of at least 1433 mils in elevation and 280 mils in depression (9720 as read on the elevation counter) with the correction counter indicating "00 mils."

3.8.2 Correction counter excursion range. The correction counter limit stops shall have a minimum of 95 mils excursion, but not over 99 mils excursion on the "+" and "-" dials.

3.8.3 Elevation accuracy. The error at any elevation setting shall not exceed ± 0.5 mil. Nine readings shall be taken in 200 mil increments. The total combined error of all readings shall not exceed 1.0 mil.

3.8.4 Backlash. The backlash in the elevating mechanism shall not exceed ± 0.7 mils at any elevation setting between 280 mils in depression (9720 as read on the elevation counter) and 1433 mils in elevation.

3.8.5 Cross level vial bubble. The cross level vial bubble shall remain centered within the thickness of a graduation line when the quadrant is elevated from 280 mils depression (9720 as read on the elevation counter) to 1433 mils elevation and the elevation level vial bubble is centered.

3.8.6 Correction counter setting effect on elevation counter. Setting a positive reading into the correction counter shall decrease the elevation counter reading by the same amount within ± 0.5 mils. Setting a negative reading into the correction counter shall increase the elevation counter reading by the same amount within ± 0.5 mils.

3.8.7 Correction counter setting effect on elevation level bubble. Any change in position of the elevation level vial bubble shall not be greater that the thickness of a graduation line when any value within the excursion range of the correction counter is set.

3.9 Knob torque. The torque for the elevation and correction knobs shall be within the ranges shown in Table II at the specified temperatures.

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TABLE II. Knob torque.

Knob	Torque at +60°F to +90°F	Torque at +145 ± 5°F	Torque at -50 ± 5°F
Elevation	3 to 10 inch pounds	13 inch pounds maximum	13 inch pounds maximum
Correction	1 to 4 inch pounds	8 inch pounds maximum	8 inch pounds maximum

3.10 Counter numerals. With the unit digit of each counter centered in the aperture of the cover, the remaining numbers shall be in alignment within 1/16th inch.

3.11 Level vial illumination. With fresh batteries installed and the light turned on, the level vial bubbles and markings shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into night.

3.12 Counter illumination. With fresh batteries installed and the light turned on, the numerals on the elevation and correction counters shall be clearly distinguishable when observed in ambient light conditions ranging from dusk into night.

3.13 Interchangeability. The quadrant shall be able to be installed on any M172A1 mount that is within specification.

3.14 Workmanship. Workmanship of the quadrant shall be in accordance with the requirements of MIL-F-13926.

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4. VERIFICATION

TABLE III. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 2 - Demonstration 3 - Examination 4 – Test						Class of Verification A - First Article B - Conformance		
Section 3 Requirements	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1	First article			X	X	X		4.2
3.2	Conformance			X	X		X	4.3
3.3	Quadrant				X	X	X	4.3.2.1
3.4.1	Storage temperature				X	X	X	4.4.1
3.4.2	Operating temperature				X	X	X	4.4.2
3.4.3	Shock				X	X	X	4.4.3
3.4.4	Vibration “A”				X	X		4.4.4
3.4.5	Vibration “B”				X		X	4.4.5
3.5	Sealing				X	X	X	4.5
3.6	Purging				X	X	X	4.6
3.7	Cleanliness				X	X	X	4.7
3.8.1	Elevation counter excursion range				X	X	X	4.8.1
3.8.2	Correction counter excursion range				X	X	X	4.8.2
3.8.3	Elevation accuracy				X	X	X	4.8.3
3.8.4	Backlash				X	X	X	4.8.4
3.8.5	Cross level vial bubble			X		X	X	4.8.5
3.8.6	Correction counter setting effect on elevation counter				X	X	X	4.8.6
3.8.7	Correction counter setting effect on elevation level bubble				X	X	X	4.8.7
3.9	Knob torque				X	X	X	4.9
3.10	Counter numerals			X		X	X	4.10
3.11	Level vial illumination			X		X	X	4.11
3.12	Counter illumination			X		X	X	4.12
3.13	Interchangeability				X	X	X	4.13
3.14	Workmanship			X		X	X	4.14

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4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article. When specified, a sample shall be subjected to first article verification in accordance with Table IV.

TABLE IV. First article inspection.

Examination or Test	Section 3 Requirement	Section 4 Verification
Quadrant	3.3	4.3.2.1
Storage temperature	3.4.1	4.4.1
Operating temperature	3.4.2	4.4.2
Shock	3.4.3	4.4.3
Vibration “A”	3.4.4	4.4.4
Sealing	3.5	4.5
Purging	3.6	4.6
Cleanliness	3.7	4.7
Elevation counter excursion range	3.8.1	4.8.1
Correction counter excursion range	3.8.2	4.8.2
Elevation accuracy	3.8.3	4.8.3
Backlash	3.8.4	4.8.4
Cross level vial bubble	3.8.5	4.8.5
Correction counter setting effect on elevation counter	3.8.6	4.8.6
Correction counter setting effect on elevation level bubble	3.8.7	4.8.7
Knob torque	3.9	4.9
Counter numerals	3.10	4.10
Level vial illumination	3.11	4.11
Counter illumination	3.12	4.12
Interchangeability	3.13	4.13
Workmanship	3.14	4.14

4.2.1 First article quantity. First article inspections shall be performed on three (3) complete units.

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4.2.2 First article inspection to be performed. The first article inspection shall be performed in accordance with Table IV.

4.2.3 First article rejection. If any item of the sample fails to comply with any of the first article requirements, the first article shall be rejected.

4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with the lot formation requirements of MIL-STD-1916, paragraph "Formation and identification of lots and batches."

4.3.2 Conformance inspection quantity. Conformance inspection quantities shall be in accordance with MIL-STD-1916, paragraph "Sampling of lots or batches." Conformance criteria shall be in accordance with 4.3.2.1 and MIL-STD-1916, paragraph "Verification level (VL)."

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4.3.2.1	Quadrant.			Drawing number 13005102
Classification	Examination or test	Conformance criteria	Requirement paragraph	Next higher assembly None Inspection method reference
Critical	None defined			
Major				
101	Presence of Plate, Identification, 13010955	100%	3.3	Visual
102	Completeness of assembly	100%	3.3	Visual/tactile
103	Storage temperature	3 units per lot 1/	3.4.1	4.4.1
104	Operating temperature	100%	3.4.2	4.4.2
105	Shock	100%	3.4.3	4.4.3
106	Vibration “B”	100%	3.4.5	4.4.5
107	Sealing	100%	3.5	4.5
108	Purging	100%	3.6	4.6
109	Cleanliness	100%	3.7	4.7
110	Elevation counter excursion range	100%	3.8.1	4.8.1
111	Correction counter excursion range	100%	3.8.2	4.8.2
112	Elevation accuracy	100%	3.8.3	4.8.3
113	Backlash	100%	3.8.4	4.8.4
114	Cross level vial bubble	100%	3.8.5	4.8.5
115	Correction counter setting effect on elevation counter	100%	3.8.6	4.8.6
116	Correction counter setting effect on elevation level bubble	100%	3.8.7	4.8.7
117	Knob torque	100%	3.9	4.9
118	Counter numerals	100%	3.10	4.10
119	Level vial illumination	100%	3.11	4.11
120	Counter illumination	100%	3.12	4.12
121	Interchangeability	100%	3.13	4.13
Minor				
201	Workmanship	VL-II	3.14	4.14

NOTES:
1/ Lot shall be accepted if there are 0 failures, and rejected if there are 1 or more failures.

4.3.3 Conformance inspection to be performed. The conformance inspection shall be performed in accordance with Table III and 4.3.2.1.

4.3.4 Conformance rejection. If any item of the sample fails to comply with any of the conformance requirements, the lot shall be rejected.

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4.4 Environmental.

4.4.1 Storage temperature. With fresh batteries installed and quadrant purged with dry nitrogen per 3.6, the quadrant shall be placed in the test chamber and the temperature of the chamber reduced gradually to -60 ± 5°F and shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, the quadrant shall be removed from the environmental chamber and inspected for signs of physical failure or damage. The quadrant shall be placed back in the environmental chamber and the temperature shall then be gradually raised to +160 ± 5°F and the quadrant shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, the quadrant shall be removed from the environmental chamber and inspected for signs of physical failure or damage. The quadrant shall be placed back in the environmental chamber and the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60°F to +90°F) at which time the quadrant shall be removed from the test chamber. The quadrant shall then meet the requirements of 3.5 through 3.14 inclusive. See 6.6, Thermal shock.

4.4.2 Operating temperature. With fresh batteries installed and quadrant purged with dry nitrogen per 3.6, the quadrant shall be placed in the environmental chamber and the temperature of the chamber reduced gradually to -50 ± 5°F. The quadrant shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period, and while at -50 ± 5°F, the quadrant shall meet requirements of 3.8 through 3.14 inclusive. Upon completion of this portion of the test, the quadrant shall then be placed back in the environmental chamber and the temperature shall be raised gradually to +145 ± 5°F. The quadrant shall remain at this temperature for a period of 4 hours. At the completion of this 4 hour period, and while at +145 ± 5°F the quadrant shall again meet requirements of 3.8 through 3.14 inclusive. Upon completion of this test, the quadrant shall be returned to the environmental chamber and the temperature shall be gradually reduced to room ambient temperature (+60°F to +90°F) at which time the quadrant shall be removed from the test chamber and shall meet the requirements as specified in 3.5 through 3.14 inclusive. See 6.6, Thermal shock.

4.4.3 Shock. The quadrant shall be mounted utilizing the keyway and four mounting bolts torqued to 25-30 foot-pounds; fixture 11747195 may be used to adapt the quadrant to the shock test equipment. The quadrant shall be subjected to the shock requirements in accordance with 3.4.3. Upon completion of this test, the quadrant shall be removed from the shock test equipment and shall meet the requirements as specified in 3.5 through 3.14 inclusive.

4.4.4 Vibration “A”. The quadrant shall be mounted utilizing the keyway and four mounting bolts torqued to 25-30 foot-pounds; fixture 11747195 may be used to adapt the quadrant to the vibration test equipment. The quadrant shall be subjected to the vibration requirements in accordance with 3.4.4. Upon completion of this test, the quadrant shall be removed from the vibration test equipment and shall meet the requirements as specified in 3.5 through 3.14 inclusive.

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4.4.5 Vibration "B". The quadrant shall be mounted utilizing the keyway and four mounting bolts torqued to 25-30 foot-pounds; fixture 11747195 may be used to adapt the quadrant to the vibration test equipment. The quadrant shall be subjected to the vibration requirements in accordance with 3.4.5. Upon completion of this test, the quadrant shall be removed from the vibration test equipment and shall meet the requirements as specified in 3.5 through 3.14 inclusive.

4.5 Sealing. Prior to the start of the test, secure the poppet valve in a closed position. Pressurize the quadrant with dry nitrogen to an internal pressure of 5.0 ± 0.10 psig, and then remove the pressure source. After a minimum time of 1 hour, check the internal pressure of the quadrant. The internal pressure must be within the range specified.

4.6 Purging. Connect the test equipment to the purging valve of the quadrant. Adjust the dry nitrogen pressure to 8.00 ± 0.25 psig. A dew point tester (standard type shop equipment) shall be used to measure the nitrogen dew point for conformance with 3.6. Disconnect the purging apparatus when the relief valve has reseated to maintain the internal pressure of 1.0 ± 0.1 psig. Purging is to be performed at standard ambient temperature and prior to performing any temperature test.

4.7 Cleanliness. Visually inspect for defects. Internal inspection shall be made by viewing through the windows.

4.8 Performance. Verifications 4.8.1 through 4.8.7 shall be done with the starting position of the quadrant as follows:

a. The flat mounting surface of the quadrant in a vertical plane.

b. The locating keyways in a horizontal plane.

c. The elevation level vial bubble centered within the thickness of a graduation line.

d. The four quadrant mounting bolts torqued to 25 to 30 foot pounds.

e. The correction counter indicating "00 mils."

f. The elevation counter indicating "0000 mils", with backlash removed. To ensure that the backlash has been removed, turn the knob clockwise to reach the desired elevation setting; if turning the knob counterclockwise, over travel the desired setting by at least a half turn of the knob and then come back to it clockwise.

4.8.1 Elevation counter excursion range. Mount the quadrant on a cross level fixture in the orientation specified in 4.8. Manually rotate the elevation counter knob until the elevation counter has traveled the minimum elevation and depression specified as read on the elevation counter.

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4.8.2 Correction counter excursion range. Mount the quadrant on a cross level fixture in the orientation specified in 4.8. Manually rotate the correction counter knob until the full excursion has traveled the minimum and maximum limits specified as read on the "+" and "-" dials of the correction counter.

4.8.3 Elevation accuracy. Mount the quadrant on a cross level fixture in the orientation specified in 4.8. Use the elevation knob to set the elevation counter to indicate 200 mils depression ("9800" as read on the elevation counter) with backlash removed and the correction counter indicating 00 mils. Elevate the cross level fixture until the elevation level vial bubble of the quadrant under testing is centered within the thickness of a graduation line. Do not overtravel the center position. Measure and record the angular elevation setting of the cross level fixture. Repeat the above procedure with the elevation counter set to indicate 0000 mils, 200 mils, 400 mils, 600 mils, 800 mils, 1000 mils, 1200 mils, and 1400 mils elevation. The recorded readings shall indicate the elevation counter settings within the values specified.

4.8.4 Backlash. Mount the quadrant on a cross level fixture in the orientation specified in 4.8. Ensure the elevation reading is "0000 mils" and that the backlash has been removed by turning the knob clockwise to reach the desired elevation setting; if turning the knob counter clockwise, overtravel the desired setting by at least a half turn of the knob and come back to it clockwise. Adjust the cross level fixture elevation to center the quadrant level vial bubble within the width of a level vial graduation line. Measure and observe the elevation setting of the cross level fixture. Rotate the elevation knob a minimum of one turn clockwise, and then reset the elevation counter to "0000 mils" by turning the knob counter clockwise without overtravel of the "0000 mils" setting. Adjust the cross level fixture elevation to center the quadrant level vial bubble within the width of a level vial graduation line. Measure and record the elevation setting of the cross level fixture. Repeat the above procedure with the elevation counter set to indicate 0000 mils, 200 mils, 400 mils, 600 mils, 800 mils, 1000 mils, 1200 mils, and 1400 mils elevation. The backlash is the difference between the two recorded cross level elevations at each of the quadrant elevation counter readings, and shall meet the requirements specified.

4.8.5 Cross level vial bubble. Observe the position of the cross level vial bubble of the quadrant at the elevations checked in 4.8.3. The position of the cross level vial bubble of the quadrant shall conform to the requirements specified when observed at these elevations. This test may be done concurrently with 4.8.3.

4.8.6 Correction counter setting effect on elevation counter. Mount the quadrant on a cross level fixture in the orientation specified in 4.8. Use the elevation knob to set the elevation counter to indicate 200 mils depression ("9800" as read on the elevation counter) with backlash removed and the correction counter indicating 00 mils. Elevate the cross level fixture until the elevation level vial bubble of the quadrant under testing is centered within the thickness of a graduation line. Rotate the correction counter knob to indicate "+50" on the correction counter. Observe and record the elevation counter reading. The elevation counter reading shall decrease by 50 mils within the limits specified. Rotate the

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correction counter knob to indicate "-50" on the correction counter. Observe the elevation counter reading. The elevation counter reading shall increase by 50 mils within the limits of 3.8.6. Repeat the above procedure with the elevation counter set to indicate 0000 mils, 200 mils, 400 mils, 600 mils, 800 mils, 1000 mils, 1200 mils, and 1400 mils elevation.

4.8.7 Correction counter setting effect on elevation level bubble. Observe the position of the elevation level vial bubble of the quadrant at the positive and negative corrections checked in 4.8.6. The position of the elevation level vial bubble of the quadrant shall conform to the requirements specified when observed at these corrections. This test may be done concurrently with 4.8.6.

4.9 Knob torque. The torque at standard ambient and the extreme operating temperatures of the elevation and correction counter knobs shall be determined with a standard torque measuring device utilizing appropriate adapters to fit the respective knobs. The measured torque values shall be within the limits specified.

4.10 Counter numerals. At elevations of 200 mils depression ("9800" as read on the elevation counter), 0000 mils, 200 mils, 400 mils, 600 mils, 800 mils, 1000 mils, 1200 mils, and 1400 mils, all readings shall be centered and a standard measuring scale shall be used to determine numeral (digit) alignment within the linear value specified.

4.11 Level vial illumination. Place the instrument in a dark room. The observer must become dark adapted. When this has been accomplished, switch on the quadrant light and observe the level vial bubbles and markings at various light levels that simulate dusk into darkness. These observations must meet the requirements specified.

4.12 Counter illumination. This test may be done concurrently with 4.11. Place the instrument in a dark room. The observer must become dark adapted. When this has been accomplished, switch on the quadrant light and observe the counter numerals and numeral alignment at various light levels that simulate dusk into darkness. These observations must meet the requirements specified.

4.13 Interchangeability. Quadrant interchangeability shall be verified by insertion, seating, and subsequent removal of the maximum fit gage, 11747969, into the quadrant's mating surface without binding.

4.14 Workmanship. Workmanship of the quadrant shall be confirmed in accordance with the verifications of MIL-F-13926.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military

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Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Quadrant, Fire Control M18A1, when attached to a telescope and quadrant mount, is used for laying the artillery weapon in elevation for indirect fire.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of an applicable level of preservation, packaging, and packing in accordance with MIL-STD-2073-1, Department of Defense Standard Practice for Military Packaging.

c. Packaging Data Sheet SPI 13005102 as applicable (See 6.5).

d. Applicable stock number.

e. Provisions for first article testing.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection must be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: RDAR-QEW-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1433 in the contract. When the contractor submits inspection equipment designs to the Government for approval, he must give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc).

c. The designs remaining to be submitted and the expected date of submittal.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research Development and Engineering Center (ARDEC) Drawings may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Packaging data sheet drawings. The following packaging data sheet drawings should be obtained from U.S. Army ARDEC, ATTN: RDAR-EIL-P B455, Picatinny Arsenal, NJ 07806-5000.

SPI 13005102 - Special Packaging Instruction for M18A1 Quadrant

6.6 Thermal shock. Caution should be exercised during environmental testing to avoid subjecting the quadrant to thermal shock. A rate of change of 5°F per minute has been shown not to thermally shock the units; however a different rate of temperature change may be used at the tester's discretion.

6.7 Subject term (key word) listing.

Towed Artillery
Self Propelled Artillery
Howitzer
M777
M198

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodian:
Army-AR
Navy-OS
Air Force- 84

Preparing activity:
Army-AR
(1290-2012-002)

Review activities:
Army- AV, MI
Navy-AS
Air Force-99
DLA-CC

Civil agency:
GSA-FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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